

Particulate Matter Monitoring & Health Effects

Issue 2026-09-05 | Entry window 5 September 2026 | 7 records screened in scope

7

RECORDS IN SCOPE
(22 EXCLUDED)

0

PubMed [EDAT] HITS
ON EITHER AXIS

5

OF 7 ON THE SENSING /
EXPOSURE AXIS

2

WITH A HUMAN
HEALTH ENDPOINT

2

METADATA ONLY
(NO ABSTRACT)

Signal of the day

Aviation and roadway leave separable fingerprints on the same neighbourhood — but only if the network measures particle *number* and absorption, not mass.

Sawan et al. (*Environ Sci Technol*) ran a hybrid mobile and fixed-site campaign in a marginalized northwest Toronto neighbourhood sitting downwind of Toronto Pearson and a freight corridor, and showed the two dominant combustion sources decouple spatially. Ultrafine particle number tracked flight activity ($r = 0.58$, $p < 0.05$ at 3.5 km) with 1.7–3.3-fold enrichment under downwind conditions, still 3.2-fold at residential sites ~8 km from the airport; black carbon instead followed roadway proximity and freight, correlating weakly with flights. The diagnostic detail is that BC–UFP correlation collapses from $r = 0.62$ under northerly winds to $r = 0.1$ downwind of the airport — shared sources versus independent ones, visible only because both pollutants were measured together. Random Forest models ($R^2 = 0.79$ UFP, 0.87 BC) separated airport proximity as a UFP-specific driver from freight trucking as a BC-specific one. For distributed low-cost networks the implication is unwelcome and concrete: a PM_{2.5}-mass node sited here would have reported an essentially null aviation signal, because the aviation contribution lives in a size range that contributes negligible mass. Source separation of this kind needs a CPC-class number channel and an absorption channel at the same node, which is precisely the pairing most low-cost deployments omit.

What changed today

- **PubMed returned a true zero, verified two ways.** No record entering PubMed on 5 September matched either the health axis or the sensing axis. This is not a harvester fault: 544 records entered PubMed that day against 4,765 on Friday 4 September, and the independent PubMed connector run with looser field tags also returned 0/0. Saturday deposits are thin and, this week, contained no PM work at all.
- **The issue was carried entirely by the non-PubMed legs.** Crossref-by-ISSN supplied four of seven records and Europe PMC three. Europe PMC returned 11 hits but **seven were already shipped in the 4 September issue** — its CREATION_DATE runs about a day behind PubMed [EDAT], so cross-source deduplication, not recall, is what that leg mostly exercises.
- **Instrumentation dominates a health-thin day.** Five of seven records sit on the sensing or exposure-assessment axis (71%, second only to 20 August's 78%), and only **two records carry a human health endpoint at all**. No record in the window reports an effect estimate with a confidence interval, so there is no forest plot today — the seventh such issue of 42.
- **Two source-attribution results worth reading together.** Sawan et al. separate aviation UFP from roadway BC by wind sector; Shi S et al. show that vanadium, the canonical ship-plume tracer, is carried by only **35.8% of plume particles** and that coverage moves with vessel speed (29.5% at 4.9 knots vs. 39.7% at 13.4). Both are warnings that a single-species or single-metric tracer under-identifies its own source.
- **A speed-reduction policy may trade CO₂ for composition.** In Shi S et al.'s matched-vessel pair the CO₂ benefit of slowing coincided with a carbon-rich particle shift and lower detectable-vanadium coverage. They also find **~20% of analysed particles meet single-particle biomass-burning criteria**, i.e. ship plumes generate false positives in exactly the source-apportionment schemes used to police them.
- **Trial watch is a structural zero, not a topical one.** ClinicalTrials.gov posted *no* updates registry-wide on 5 September — nor on 29/30 August — against 973–1,001 on each weekday. Every Saturday and Sunday issue will carry this zero for calendar reasons, and it should not be read as an absence of PM trial activity.
- **Two records ship without abstracts.** The *Atmos Pollut Res* tire-wear tracer review and the *Atmos Res* dust-aging lidar paper are indexed by Crossref with no abstract and no Europe PMC record; both are titled informatively enough to place but not to assess, and both go on the retry list.

Corpus analytics

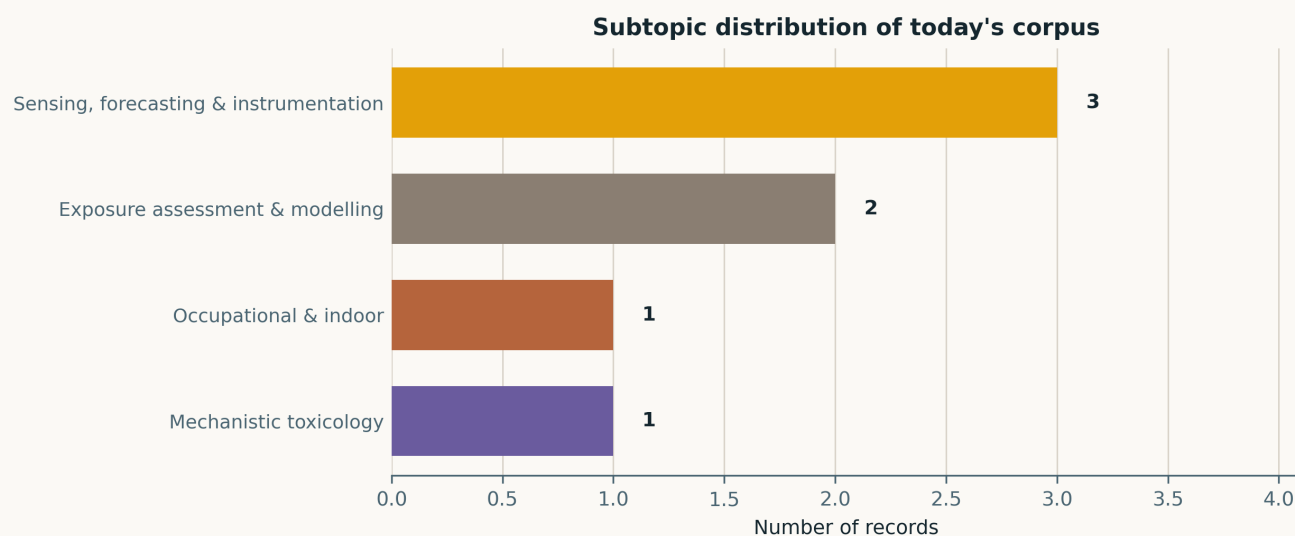


Fig. Subtopic assignment of the seven in-scope records. Four of the ten clusters are occupied — tying the fewest recorded (5, 23 and 24 August) — and at seven records this is the **smallest single-day issue this watch has shipped**, below the previous minimum of eight on 29 August. *Sensing, forecasting & instrumentation* leads with three; six of the ten clusters, including all four of the cardiovascular, neuro, respiratory and reproductive health blocks, are empty.

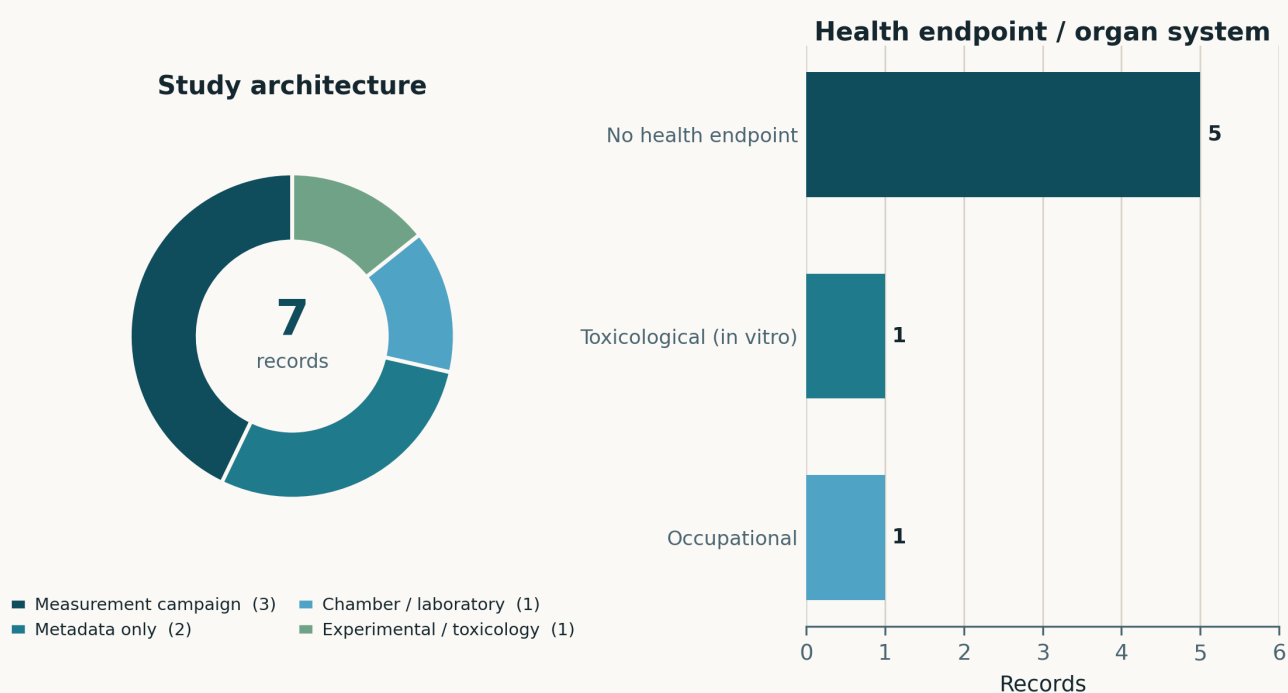


Fig. Left — study architecture. *Measurement campaign* accounts for three of seven, with one laboratory-combustion study, one in-vitro toxicology preprint and two records that could not be classified beyond *Metadata only* because no abstract has been deposited. Right — health endpoint. Five records carry no human endpoint at all (two aerosol chemistry, one instrument, one exposure, one with no health endpoint), which is why the endpoint panel is so sparse; only the Johannesburg occupational study and the in-vitro preprint sit on a health axis.

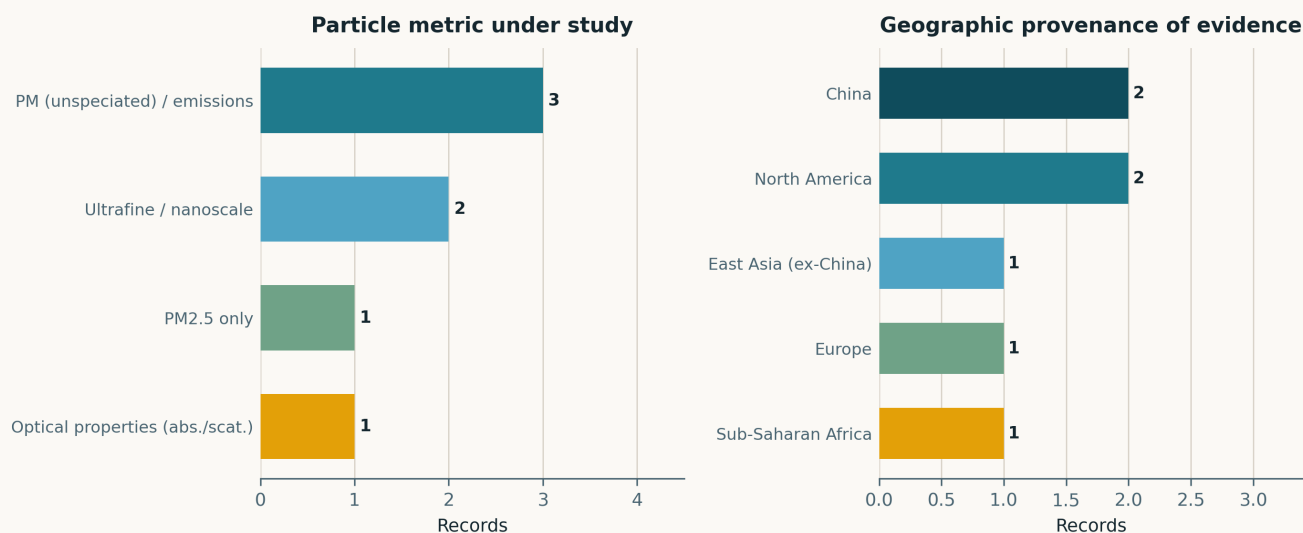


Fig. Left — particle metric. Four metrics span seven records: ultrafine number (2), unspicated PM and emission factors (3), lidar optical properties (1) and PM_{2.5} (1). **PM_{2.5} mass appears alone exactly once, and only as a dosed in-vitro concentration** (25–100 µg/mL) rather than an ambient measurement — every ambient record here is instrumented on number, emission factors or optical properties instead. Right — geographic provenance. North America and China contribute two each; the Johannesburg occupational study is the day's only sub-Saharan African record, and South Asia, Latin America and Oceania are absent.

Life-course windows addressed today (records may span several stages)

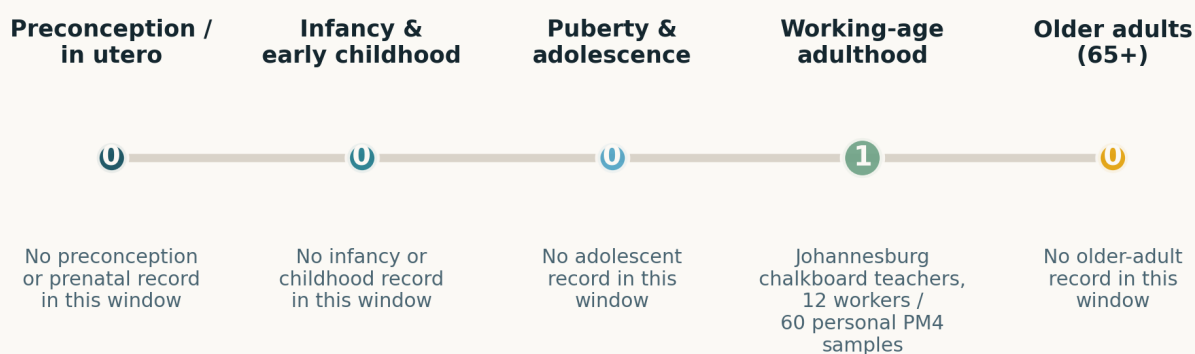


Fig. Life-course windows addressed. Only the working-age stage is occupied, by the 12-teacher Johannesburg personal-sampling study; the preconception, infancy, adolescence and 65+ windows are all empty. A single-stage day is a direct consequence of a corpus that is 71% instrumentation and exposure assessment.

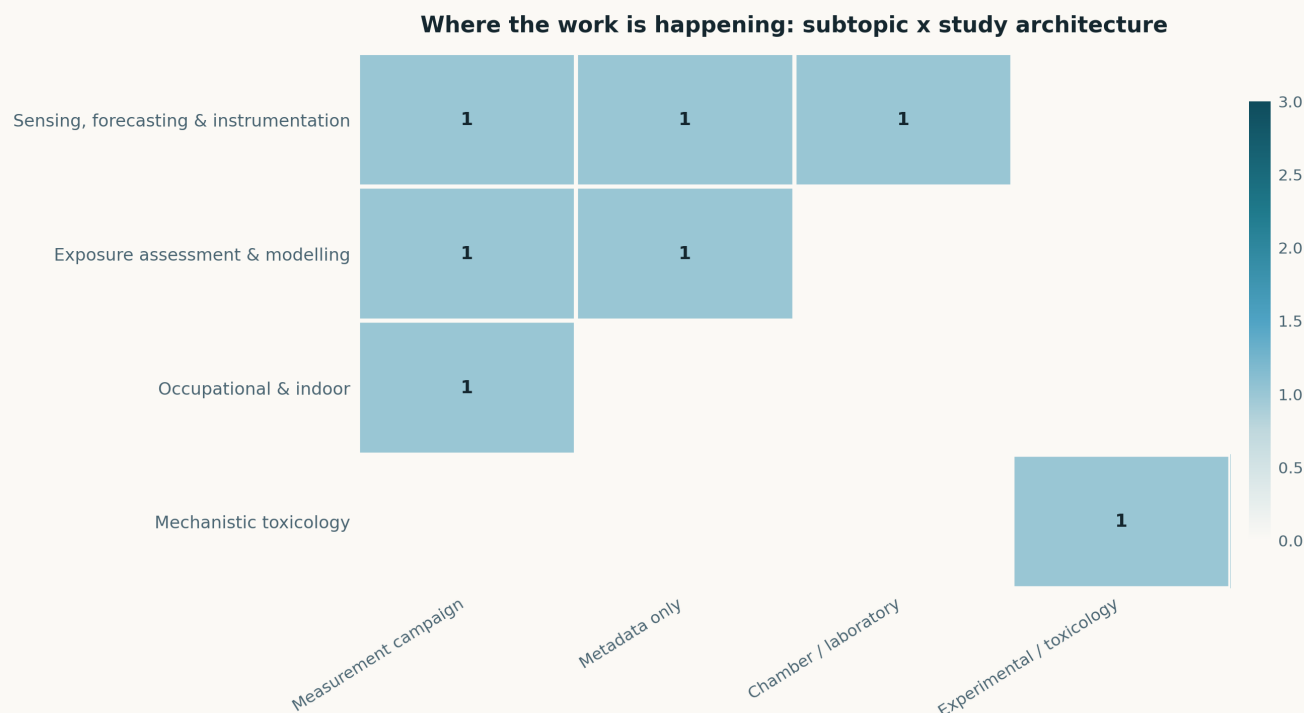


Fig. Subtopic × study-architecture cross-tabulation. With seven records the matrix is necessarily sparse, but the standing structural finding survives at small sample: the *Sensing* and *Exposure* rows again contain *no* health-endpoint linkage, and the one mechanistic record is again purely experimental with no bridge to human biomarker data. There is no forest plot in this issue — no record in the window reports an effect estimate with a confidence interval.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

3 records

Shi S et al. 2026 npj Clim Atmos Sci

Vessel speed reduction trades CO₂ against plume composition — and against the tracer that identifies the plume

UAV interception of 14 fresh ship plumes with single-particle mass spectrometry and population-balance inversion to reconstruct ultrafine and coarse modes. Two marker gaps emerge. **Only 35.8% of the reconstructed particle-number population carried detectable vanadium** (29.5% at 4.9 knots vs. 39.7% at 13.4 knots in a matched-vessel pair; 0–70.8% across a speed-blind manifold), so the canonical ship tracer is operation-dependent rather than conserved. Separately, **~20% of analysed particles met biomass-burning criteria**, a false-positive channel in ship-influenced samples. Sulfate-related ion signals ran 2.3–10.7-fold higher in V/VO-bearing particles. The authors are careful that laser desorption/ionization matrix effects may contribute to that contrast, which is the right caveat and limits how far the V–S co-location can be pushed mechanistically. Single campaign, 14 plumes, one matched pair — the speed comparison rests on very few vessels.

doi: 10.1038/s41612-026-01532-3

Wang S et al. 2026 Fire Saf J

Smoldering chaparral emits 10–70× the particulate of flaming, and fuel moisture sets the multiplier

FTIR quantification of gas and particulate emissions from controlled flaming and smoldering combustion of four dominant California chaparral species at varying fuel moisture. **Smoldering produced 10–70-fold higher PM emission factors than flaming across all species**, with the amplification ratio itself varying by fuel type and moisture; higher moisture enhanced incomplete-combustion products, principally CO and hydrocarbons, with live eucalyptus highest at 37.18 ± 0.65 g/kg hydrocarbons under smoldering. Cross-ecosystem comparison found chaparral broadly similar to western shrubland and grassland fuels except in nitrogen-containing species. The limitation that matters is scale: these are bench-scale burns, and the authors note combustion scale and mixing conditions shift emission profiles, so transfer to landscape fire inventories is an assumption rather than a result. Directly useful for anyone weighting wildfire smoke exposure by combustion phase rather than by total burned area.

doi: 10.1016/j.firesaf.2026.104687

Ji et al. 2026 Atmos Res

Dust aerosol aging tracked by combined Mie, polarization and fluorescence lidar

Metadata only. Crossref carries the title, author list (8) and 5 September registration date but no abstract, and there is no Europe PMC record; the publisher's full text was not retrievable on this run. From the title the study links optical evolution during dust aging to chemical transformation using a three-channel lidar — a combination relevant to anyone attempting composition inference from optical instruments rather than mass. **No quantities, site, campaign length or retrieval performance are asserted here, because none could be verified.** Added to the abstract retry list.

doi: 10.1016/j.atmosres.2026.109312

Exposure assessment & modelling

2 records

Sawan et al. 2026 Environ Sci Technol*Aviation drives ultrafine number, freight drives black carbon, in the same neighbourhood*

Hybrid mobile and fixed-site campaign in a marginalized northwest Toronto neighbourhood downwind of Toronto Pearson and a freight corridor. UFP correlated with flight activity ($r = 0.58$, $p < 0.05$ **at 3.5 km**) and was enriched **1.7–3.3-fold** downwind of the airport, measured as the ratio of interquartile ranges under downwind versus other wind conditions, reaching **3.2-fold at residential sites ~8 km away**. BC instead tracked roadway proximity and freight and correlated weakly with flight counts. BC–UFP correlation was strong under northerly winds (mean $r = 0.62$) but collapsed to $r = 0.1$ downwind of the airport. Random Forest models ($R^2 = 0.79$ UFP, 0.87 BC) identified industrial land use and a highway as shared drivers, airport proximity as UFP-specific and freight trucking as BC-specific. The inference is observational and wind-sector stratified, so airport attribution rests on a co-location argument rather than on any tracer unique to jet exhaust. The lesson for network design is that mass-based siting would have missed the aviation signal entirely.

doi: 10.1021/acs.est.6c06666

Scalabrin et al. 2026 Atmos Pollut Res*Chemical tracers for tire wear particles in atmospheric aerosol: analytical approaches and challenges*

Metadata only. Crossref registers the title, 11 authors and a 5 September creation date with no abstract, and the record has no Europe PMC counterpart. The title places it as a review of tracer chemistry for apportioning non-exhaust tire wear — a live problem, since tire wear is a growing share of traffic PM precisely as exhaust controls tighten, and the standard markers each fail differently. **No findings, recommended tracer or study count are asserted here.** A targeted Scholar Gateway full-text query returned only adjacent June 2026 reviews, not this paper. Added to the abstract retry list.

doi: 10.1016/j.j.apr.2026.103194

Occupational & indoor exposure**1 record****Mbazima et al. 2026** Int J Environ Health Res*Chalk dust puts teachers over a deposition threshold while staying under the occupational limit*

Sixty personal breathing-zone samples from 12 Johannesburg secondary-school teachers (7 female, 5 male), collected Monday–Friday 08:00–14:35 with GilAir-3 pumps, a $4\ \mu\text{m}$ cut-point cyclone and 25 mm PVC filters at $1.85\ \text{L}/\text{min}$, then run through a multiple-path particle dosimetry model at low, moderate and high exertion for nasal and oronasal breathing. **PM_{10} ranged $0.019\text{--}0.761\ \text{mg}/\text{m}^3$, mean $0.126 \pm 0.134\ \text{mg}/\text{m}^3$** ; chalk use ($\beta = 0.468$) and sampling duration ($\beta = 0.305$, both $p < 0.001$) predicted concentration. Breathing route dominated deposition, and alveolar clearance was slower than tracheobronchial in both sexes. The authors' conclusion — that risk persists below the occupational exposure limit — rests on modelled rather than measured dose, and 12 teachers in one region is a small and non-random base. Note the standard deviation exceeds the mean, so the distribution is heavily right-skewed and the mean is a poor summary.

doi: 10.1080/09603123.2026.2726420

Mechanistic toxicology**1 record****Han G et al. 2026 (preprint)** Research Square (preprint)*Picropodophyllotoxin attenuates $\text{PM}_{2.5}$ injury in pulmonary endothelial cells via mTOR-dependent autophagy*

Human pulmonary artery endothelial cells exposed to $\text{PM}_{2.5}$ at $25\text{--}100\ \mu\text{g}/\text{mL}$ for 24–48 h with picropodophyllotoxin ($2\text{--}50\ \mu\text{M}$) or dexamethasone ($50\ \mu\text{M}$). $\text{PM}_{2.5}$ reduced viability dose- and time-dependently with increased LDH leakage and ROS and suppressed SOD/CAT and SGK1; PPT restored viability, SGK1 and mTOR phosphorylation and attenuated TLR4, MyD88, LC3II and Beclin 1 upregulation. **Not peer reviewed, single cell line, no animal or human data, and no dose–response statistics reported in the abstract.** The exposure concentrations are far above realistic alveolar deposition, which is the standard weakness of this design and limits the finding to pathway plausibility rather than therapeutic inference. Included for completeness of the mechanistic map, not as evidence of effect.

doi: 10.21203/rs.3.rs-10446885/v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Mechanistic toxicology					
Han G et al. 2026	Picropodophyllotoxin attenuates PM _{2.5} -induced pulmonary injury via modulation of oxidative stress and mTOR-mediated autophagy	<i>Research Square (preprint)</i>	Experimental / toxicology	PM _{2.5} dosed in vitro at 25-100 ug/mL	East Asia (ex-China)
Occupational & indoor					
Mbazima et al.	Exposure and respiratory tract deposition of particulate matter among teachers using chalkboards in selected public secondary schools in the Johannesburg region	<i>Int J Environ Health Res</i>	Measurement campaign	Respirable PM ₄ , personal breathing-zone sampling + MPPD deposition model	Sub-Saharan Africa
Sensing, forecasting & instrumentation					
Ji et al.	Quantifying the aging of dust aerosols using Mie-polarization-fluorescence lidar: linking optical evolution to chemical transformations	<i>Atmos Res</i>	Metadata only (no abstract)	Dust optical properties by Mie-polarization-fluorescence lidar	China
Shi S et al.	Potential aerosol tradeoffs of vessel speed reduction for fresh ship plume composition and tracer coverage	<i>npj Clim Atmos Sci</i>	Measurement campaign	Ultrafine and coarse modes of fresh ship plumes; V/VO and sulfate markers	China
Wang S et al.	Impact of fuel moisture on chaparral emissions under flaming and smoldering conditions	<i>Fire Saf J</i>	Chamber / laboratory	PM and gas emission factors by FTIR, flaming vs smoldering	North America
Exposure assessment & modelling					
Sawan et al.	Uncoupling aviation and roadway contributions to ultrafine particle and black carbon exposure in a marginalized neighborhood	<i>Environ Sci Technol</i>	Source apportionment	Ultrafine particle number and black carbon, wind-sector stratified	North America
Scalabrin et al.	Chemical tracers for assessing the contribution of tire wear particles in atmospheric aerosol: analytical approaches, challenges and future perspectives	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Non-exhaust tire wear particles via chemical tracers	Europe

Provenance and method

Entry window **5 September 2026**, a single day. **PubMed** (NCBI E-utilities) was queried on both the health and sensing axes against [Date - Entry] and returned **zero hits on each**; the PubMed connector, run independently with looser field tags, also returned zero, and a control query confirmed only 544 records of any kind entered PubMed that day against 4,765 the previous day. The issue therefore rests on **Crossref-by-ISSN** (18 raw hits, 4 in scope) and **Europe PMC** (CREATION_DATE, 11 raw hits, 3 in scope after seven were found already published in the 4 September issue). **arXiv** returned HTTP 200 but its newest submission remains 2026-08-20 for the third consecutive run, so it carried nothing. **OpenAlex** returned HTTP 429 (unchanged since 28 July). **Consensus** returned three hits, all verified out of window by Crossref created date (2018, 2019, 2025) — a nil for the seventh consecutive run. **ClinicalTrials.gov** posted no updates registry-wide on 5 September; the same holds for 29 and 30 August against 973–1,001 on each intervening weekday, so the weekend trial zero is a posting-calendar artefact and is reported as such. Twenty-two records were screened out with a logged reason: seven as duplicates of the 4 September issue and fifteen as off-topic or non-research (aquatic microplastics and PFAS, a corrigendum, a correspondence letter, indoor lighting, urban heat island and precipitation forecasting). Subtopic, design, particle-metric and geography labels are assigned from title, abstract and keywords; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Two records carry no abstract from any source and are labelled *Metadata only*; nothing is asserted about their findings. All DOI links resolve to the publisher of record. Abstracts remain the copyright of their respective publishers.